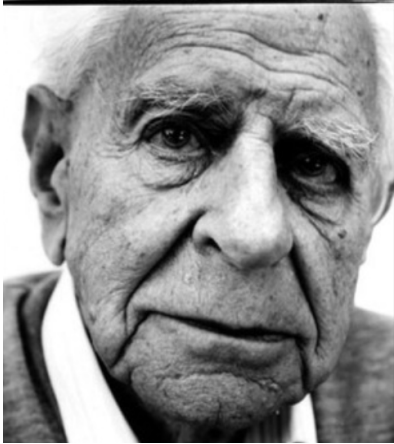




A theory is just a mathematical
model to describe the observations.

— *Karl Popper* —

Graph Theory — Worksheet

Chapter 10 • Rosen: Discrete Mathematics and Its Applications

In this worksheet you will explore:

- Representing real-world networks as graphs
- Types of graphs
- Graph terminology
- Connectivity
- Euler and Hamiltonian paths & circuits

“A graph is worth a thousand adjacency lists.”

Student Name:

Student ID:

Semester / Section:

Welcome to this lecture of Graph Theory !

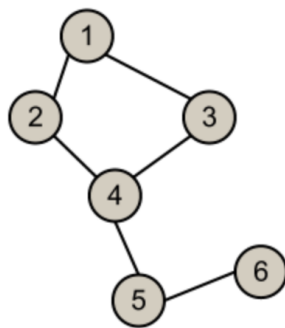
So far, we have studied *sets* and *relations*, where we learned how elements can be connected with each other. Then, in *algebraic structures*, we explored how to impose operations on sets to give them more structure. Now, we move one step further- what if we want to *visualize* and *analyze* relationships between objects! This leads us to **Graph Theory**, where sets and relations come alive through diagrams.

Formal definition of a **Graph**

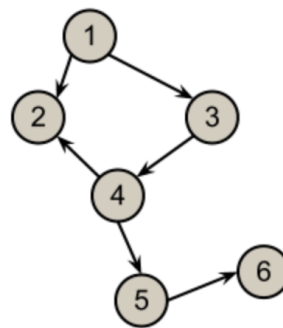
A graph $G = (V, E)$ consists of sets, V : a nonempty set of vertices (or nodes); and E : a set of edges. Each edge has either one or two vertices associated with it, called its *endpoints*. An edge is said to connect its endpoints.

Question!

Can these edges have directions??



Undirected



Directed

Formal definition of a **Directed Graph**

A directed graph (or digraph) (V, E) consists of a nonempty set of vertices V and a set of directed edges E . Each directed edge is associated with an ordered pair of vertices. The directed edge associated with the ordered pair (u, v) is said to start at u and end at v .

Let us now look at an example of a situation. Imagine a student has recently moved to a new city. Every day she travels between several important places: her home, college, gym, and the library. There is a road between Home (H) and College (C), College and Gym (G), Gym

and Home, Home and Library (L), and Library and College. She also prepared a graph for easy reference. **Perform the following activity below.**

Draw the graph with H, C, G, L. No repeated roads; no self-loops.

Does any place have more than one road to the same place? Any self-loop?

Observation:

Add a second road between Home and College.

How is this graph different from Scenario A?

Observation:

Add a road that starts and ends at G.

What is special about this road? What does it connect?

Observation:

Replace the L–C road with a one-way road from C to L.

From which place to which can we travel? Reverse possible?

Observation:

Let us revise what we have learned so far

Graph Terminology — Quick Reference

Graph Type	Edge Type	Multiple Edges?	Loops?
Simple Graph	Undirected	No	No
Multigraph	Undirected	Yes	No
Pseudograph	Undirected	Yes	Yes
Simple Directed	Directed	No	No
Directed Multi-graph	Directed	Yes	Yes
Mixed Graph	Directed & Undirected	Yes	Yes

Basic Graph Terminology

Let us learn some standard vocabulary for describing graphs - its vertices and edges.

Definition 1 (Adjacent vertices & Incident edges). *Two vertices u and v in an undirected graph G are called **adjacent** (or neighbors) in G if $\{u, v\}$ is an edge e of G . Such an edge e is called **incident** with u and v , and e is said to **connect** u and v .*

Let us build some intuition before the next definition: $N(v)$ denotes everyone directly connected to v , and $N(A)$ denotes everyone connected to at least one vertex in the set A .

Definition 2 (Neighborhood). *Let $G = (V, E)$ be a graph and $v \in V$. The **neighborhood** of v , denoted $N(v)$, is the set of all neighbors of v . For $A \subseteq V$,*

$$N(A) = \bigcup_{v \in A} N(v).$$

Example 1. *Try your hands on this example. Write the degrees and the neighborhoods of the vertices in the graphs G and H given in the figure below.*

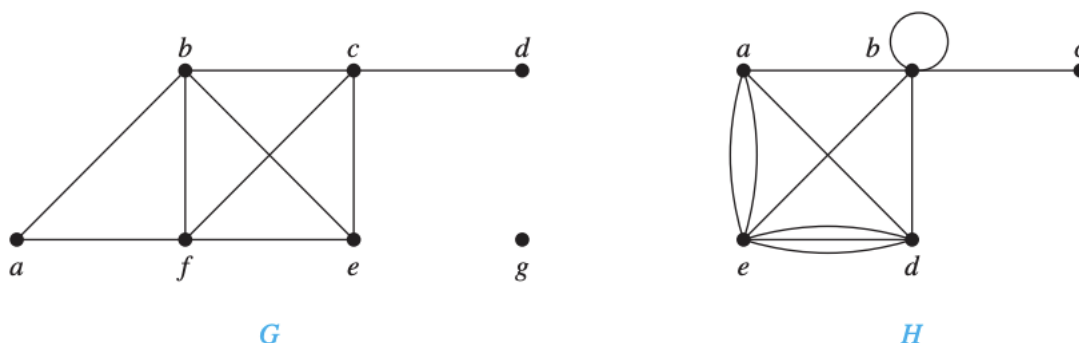


Figure 1: Undirected graphs G and H

Write the degrees and neighborhoods for vertices of G here.	Write the degrees and neighborhoods for vertices of H here.

Discovering Special Vertices in the Graphs

Look again carefully at graphs G and H . Some vertices are very “social” - connected to many others; some have very few or even no connections. Here, we have two special names for special vertices.

- A vertex with **no edges incident to it** (degree = 0) is called an **isolated vertex**.
- A vertex with **exactly one incident edge** (degree = 1) is called a **pendant vertex**.

Observation Task: Without calculating (just by *looking* at the drawing) spot these special vertices visually.

- Which vertex in graph G is completely disconnected from the rest?
- Which vertex in graph G is connected to exactly one other vertex?
- Do the same for graph H .

Isolated vertices: _____

Pendant vertices: _____

Observe the Sum of Degrees of G and H

Question: Compute the sum of degrees for both G and H . What pattern do you notice?

Your answer: _____

Yes, that is not a coincidence. Every edge touches *two* vertices, so each edge contributes **2** to the total degree count. Therefore the sum of degrees *always* equals twice the number of edges.

Well done — you have just discovered the **Handshaking Lemma** on your own!

Theorem 1 (Handshaking Lemma). *For any undirected graph $G = (V, E)$,*

$$\sum_{v \in V} \deg(v) = 2|E|.$$

Note: The lemma applies to graphs with multiple edges or loops as well.

Challenge — Can you prove this?

Theorem. *An undirected graph has an even number of vertices of odd degree.*

You have already observed that the sum of degrees is always even. Now use that observation to reason about how many vertices can have odd degree.

Task: Write your proof in the space below.

Write your proof here.

Up to now, we counted edges incident to a vertex. But when edges have **directions**, it matters whether they go **into** or **out of** a vertex.

Definition 3 (Directed adjacency and endpoints). *Let G be a graph with directed edges. For a directed edge (u, v) :*

- u is **adjacent to** v ; v is **adjacent from** u .
- u is the **initial vertex**; v is the **terminal (end) vertex**.

For a loop, the initial and terminal vertices coincide.

Definition 4 (In-degree and Out-degree). *In a directed graph, the **in-degree** of v , written $\deg^-(v)$, is the number of edges whose terminal vertex is v . The **out-degree**, $\deg^+(v)$, is the number of edges whose initial vertex is v .*

Note: *A loop at v contributes **1** to both $\deg^-(v)$ and $\deg^+(v)$.*

Example 2. *Find the in-degree and out-degree of each vertex in the directed graph G shown below (Figure 2).*

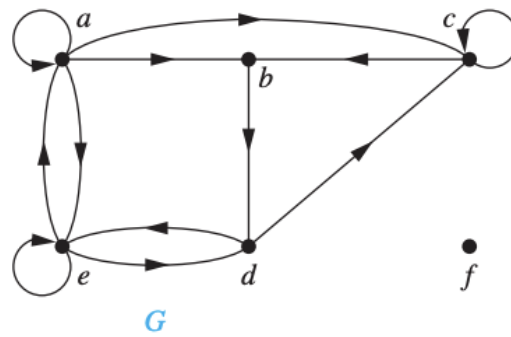


Figure 2: A directed graph

Write the degrees and neighborhoods for vertices of G here.

10.2.3 — Some Special Simple Graphs

Some families of simple graphs appear so often that they have standard names. Study the table below and fill in the thinking column.

Graph	Structure	$ E $	Why this formula?
K_n	Every pair of vertices connected by exactly one edge	$\frac{n(n-1)}{2}$	
C_n	Vertices in a single closed loop; every vertex has degree 2	n	
W_n	Cycle C_n with one hub vertex connected to all cycle vertices	$2n$	
Q_n	Vertices are n -bit strings; adjacent if they differ in exactly one bit	$n \cdot 2^{n-1}$	
$K_{m,n}$	Two vertex sets of size m and n ; every vertex on one side connects to every vertex on the other	$m \cdot n$	

10.2.4 — Bipartite Graphs

Some networks naturally divide vertices into two types - people and jobs, students and courses, users and servers — where edges only run *between* the two types, never within the same type.

Definition 5 (Bipartite Graph). *A simple graph $G = (V, E)$ is **bipartite** if V can be partitioned into two disjoint sets V_1 and V_2 such that every edge connects a vertex in V_1 to a vertex in V_2 . No edge connects two vertices within the same set.*

$$V = V_1 \cup V_2, \quad V_1 \cap V_2 = \emptyset$$

Complete Bipartite Graph $K_{m,n}$: Two vertex sets of size m and n ; every vertex on one side connects to every vertex on the other.

$$|E(K_{m,n})| = m \cdot n$$

Activity: Draw $K_{3,3}$ and C_5 below.

Answer

Quick Check: For each graph below, decide whether it is bipartite and give a short reason.

Graph	Bipartite? (Yes/No)	Reason
K_4		
C_5		
Q_3		

How to Check if a Graph is Bipartite?

A graph is bipartite if you can colour its vertices using *two colours* such that no two adjacent vertices share the same colour. This is also called **2-colourability**. Moving ahead, we have another characterisation for bipartite graphs.

Theorem 2. *A graph G is bipartite if and only if it contains no odd-length cycle.*

Proof. *Try to prove this yourself before reading further. Hint: think about what happens when you try to 2-colour a cycle of odd length.*

Write your proof attempt here.

Matchings in Bipartite Graphs

Four students S_1, S_2, S_3, S_4 are to be assigned projects P_1 (AI), P_2 (Networks), P_3 (Security), P_4 (Databases) based on their preferences:

$$S_1 : P_1, P_2 \quad S_2 : P_2, P_3 \quad S_3 : P_3, P_4 \quad S_4 : P_1, P_4$$

Each student gets exactly one project; no project goes to more than one student.

(a) Draw the bipartite graph.

Answer.

(b) Choose edges pairing students to projects with no conflicts.

Answer.

The set of non-conflicting edges you chose is called a **matching**.

Definition 6 (Matching). A **matching** in a graph is a set of edges such that no two edges share an endpoint. Each vertex appears in at most one edge of the matching.

Definition 7 (Perfect Matching). A matching is **perfect** if every vertex on the left side is matched to exactly one vertex on the right side.

Check: Did your matching in part (b) turn out to be perfect? _____

When Does a Perfect Matching Exist?

Now the preferences change:

$$S_1 : P_1 \quad S_2 : P_1 \quad S_3 : P_2, P_3 \quad S_4 : P_4$$

(c) Draw the new bipartite graph and attempt a perfect matching.

Answer.

Observation: S_1 and S_2 both want only P_1 . The group $\{S_1, S_2\}$ can access only $\{P_1\}$ — two students competing for one project. A perfect matching is impossible here. This structural bottleneck

is exactly what **Hall's Marriage Theorem** captures.

Theorem 3 (Hall's Marriage Theorem). *A bipartite graph $G = (V_1 \cup V_2, E)$ has a perfect matching if and only if for every subset $S \subseteq V_1$,*

$$|N(S)| \geq |S|.$$

That is, every group of k vertices on the left must have at least k distinct neighbours on the right.

Connectivity - Can We Reach Everywhere?

Return to the city map we saw in the beginning of this worksheet. Is it possible to travel from *any* place to *any other* place using the available roads? This idea leads to the concept of connectivity.

Definition 8 (Connected Graph). *An undirected graph is **connected** if there exists a path between every pair of distinct vertices.*

Definition 9 (Connected Component). *A **connected component** of G is a maximal connected subgraph of G . Every vertex belongs to exactly one connected component.*

Activity. Look at Figure 3. Which of G_1 and G_2 is connected? For any disconnected graph, identify and mark its connected components.

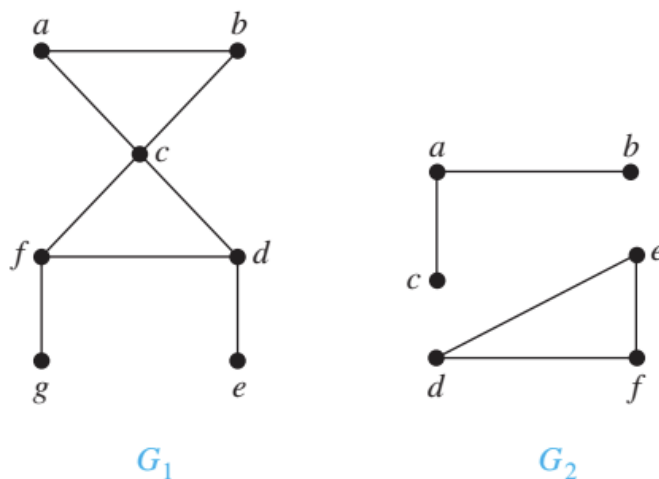


Figure 3: Graphs G_1 and G_2

Answer.

Vulnerability of a Connected Graph

A connected graph may still be *fragile*, i.e., removing just one vertex or one edge can disconnect it. This motivates the following definitions.

Definition 10 (Cut Vertex). A vertex v of a connected graph G is a **cut vertex** if removing v (and all edges incident to it) disconnects G .

Definition 11 (Cut Edge / Bridge). An edge e of a connected graph G is a **cut edge** (or **bridge**) if removing e disconnects G .

Activity. In Figure 4, identify all cut vertices and cut edges in G_1 and G_2 .

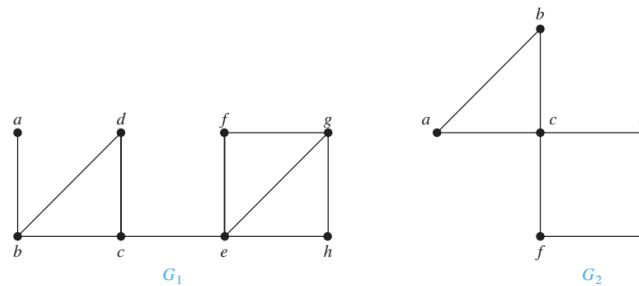


Figure 4: Find cut vertices and cut edges

Answer.

Remarks.

- A graph with a cut vertex is more vulnerable to disconnection.
- In a tree, *every* edge is a cut edge.
- A graph with no cut vertices and no cut edges has a stronger form of connectivity.

Connectivity in Directed Graphs

In directed graphs, having a path from u to v does not guarantee a path from v to u . This gives rise to two notions of connectivity.

Definition 12 (Strongly Connected). A directed graph is **strongly connected** if for every pair of vertices u and v , there is a directed path from u to v and from v to u .

Definition 13 (Weakly Connected). A directed graph is **weakly connected** if the underlying undirected graph (obtained by ignoring edge directions) is connected.

Quick Check. Circle the correct answer.

1. Every strongly connected graph is weakly connected. **True / False**
2. Every weakly connected graph is strongly connected. **True / False**

Definition 14 (Strongly Connected Component). A **strongly connected component** of a directed graph G is a maximal strongly connected subgraph of G .

Activity. Identify all strongly connected components in graph H shown in Figure 5.

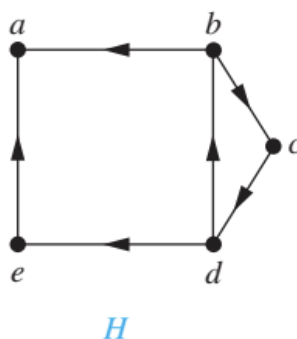


Figure 5: Find strongly connected components of H

Answer.

Euler and Hamilton Paths — Two Famous Puzzles

The Seven Bridges of Königsberg

In the 18th century, the city of Königsberg (now Kaliningrad, Russia) was built around the Pregel River, with seven bridges connecting four land masses. A puzzle arose among the citizens:

Is it possible to walk through the city, crossing each bridge exactly once, and return to your starting point?

Many people tried, but no one could do it — and no one could prove why.

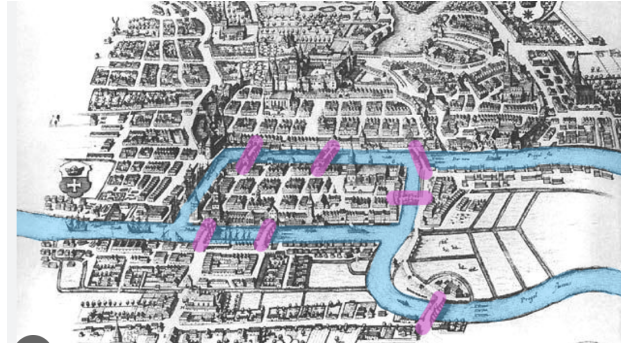


Figure 6: Seven Bridges of Königsberg

Activity 1. Draw a simple map of Königsberg with regions A (north bank), B (east bank), C (west bank), D (island) and 7 bridges. Then try to find a route that starts anywhere, crosses each bridge exactly once, and returns to the start. Record your attempts below.

<i>Map sketch and route attempts:</i> Attempt 1: _____ Attempt 2: _____ Attempt 3: _____ What difficulty did you face? _____

Euler's Brilliant Idea

In 1736, **Leonhard Euler** solved this problem, creating a whole new branch of mathematics. His key insight:

Don't think about the map. Think about the connections!

He reduced the problem to a graph: each land region = a **vertex**; each bridge = an **edge**.

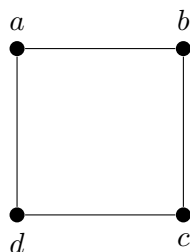
Activity 2. Draw the graph representation of Königsberg (vertices A, B, C, D with seven edges representing the bridges).

Answer.

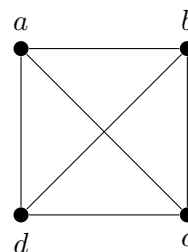
Discovering the Pattern

Activity 3. For each graph below, try to trace a path that uses every edge exactly once and returns to the starting point.

Graph 1 — A square



Graph 2 — Square with diagonals



Can you do it? _____

Can you do it? _____

Path:

Path:

Activity 4. Count the degree of each vertex for both graphs.

Graph 1 (Square):

$\deg(a) = \underline{\hspace{1cm}}$, $\deg(b) = \underline{\hspace{1cm}}$, $\deg(c) = \underline{\hspace{1cm}}$, $\deg(d) = \underline{\hspace{1cm}}$

No. of vertices with odd degree: _____

Graph 2 (Square with diagonals):

$\deg(a) = \underline{\hspace{1cm}}$, $\deg(b) = \underline{\hspace{1cm}}$, $\deg(c) = \underline{\hspace{1cm}}$, $\deg(d) = \underline{\hspace{1cm}}$

No. of vertices with odd degree: _____

Making a Conjecture.

Q1. Which graphs allowed you to trace all edges exactly once and return to the start?

Q2. What do you notice about the number of vertices with odd degree in these graphs?

Euler's Theorems

Definition 15 (Euler Circuit and Euler Path). An **Euler circuit** in a graph uses every edge exactly once and starts and ends at the same vertex. An **Euler path** uses every edge exactly once but may start and end at different vertices.

Theorem 4 (Euler's Theorems). **Theorem 1.** A connected graph has an Euler circuit if and only if every vertex has even degree.

Theorem 2. A connected graph has an Euler path (but not an Euler circuit) if and only if it has exactly two vertices of odd degree. The path must start at one odd-degree vertex and end at the other.

Why does this make sense? Every time you pass through a vertex you use two edges (one to enter, one to leave), so edges at every vertex must pair up — meaning the degree must be **even**. The start/end vertex of a non-circuit path is the only exception.

Verify with Königsberg.

$\deg(A) = ______ , \deg(B) = ______ , \deg(C) = ______ , \deg(D) = ______$

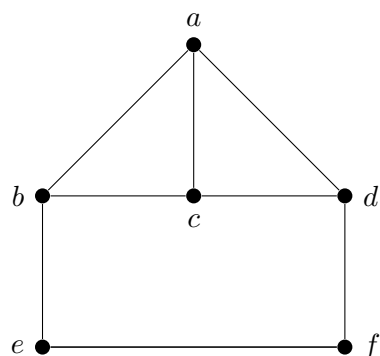
No. of vertices with odd degree: $______$

Conclusion — is the Königsberg walk possible? $______$

Why or why not? $______$

Practice — Can You Draw It Without Lifting Your Pencil?

The House graph. Determine whether the graph below has an Euler path or Euler circuit by checking degrees, then find the path if it exists.



Degrees of vertices:

$\deg(a) = \underline{\hspace{1cm}}$ $\deg(b) = \underline{\hspace{1cm}}$ $\deg(c) = \underline{\hspace{1cm}}$

$\deg(d) = \underline{\hspace{1cm}}$ $\deg(e) = \underline{\hspace{1cm}}$ $\deg(f) = \underline{\hspace{1cm}}$

No. of vertices with odd degree: $\underline{\hspace{2cm}}$

Can it be drawn without lifting pencil? $\underline{\hspace{2cm}}$

If yes, path: $\underline{\hspace{10cm}}$

A Different Kind of Journey — Hamilton's Puzzle

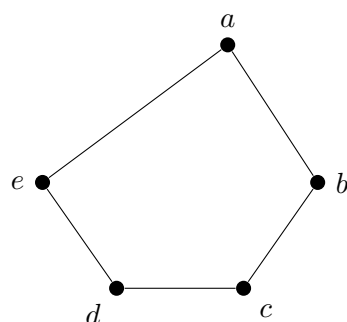
In 1857, **Sir William Rowan Hamilton** invented a different puzzle using a wooden dodecahedron with cities at its vertices.

Can you start at one city, visit every other city exactly once, and return home?

This is different from Euler's problem — here we care about visiting **vertices**, not edges.

Definition 16 (Hamilton Circuit and Hamilton Path). A **Hamilton circuit** visits every vertex exactly once and returns to the starting vertex. A **Hamilton path** visits every vertex exactly once but may start and end at different vertices.

Activity 5. Starting at vertex a in the pentagon below, find a Hamilton circuit.



Write your circuit: $\dots\dots\dots$

Euler vs. Hamilton — Comparison Table

Complete the table below.

	Euler Path / Circuit	Hamilton Path / Circuit
What do we traverse?		
What is used exactly once?		
Easy existence test?		
Real-world example		

Real-World Applications

Application 1 — The Postal Route. A mail carrier must walk down every street in a neighbourhood to deliver mail.

What kind of path is needed? _____

Why? _____

Application 2 — The Travelling Salesperson. A salesperson visits clients in 10 cities and returns home, visiting each city exactly once.

What kind of path is needed? _____

Why? _____

Challenge Problems

Challenge 1. Design a graph with 6 vertices that has an Euler circuit but *no* Hamilton circuit.

Draw your graph and explain why it satisfies both conditions.

Explanation: _____

Challenge 2. If the city planners wanted to add **one new bridge** to Königsberg to make an Euler circuit possible, where should they add it? Explain your reasoning.

Your answer and reasoning:

Sufficient Conditions for Hamilton Circuits

Although determining whether a graph has a Hamilton circuit is difficult in general, two important theorems give useful sufficient conditions.

Theorem 5 (Dirac's Theorem). *If G is a simple graph with $n \geq 3$ vertices such that $\deg(v) \geq n/2$ for every vertex v , then G has a Hamilton circuit.*

Theorem 6 (Ore's Theorem). *If G is a simple graph with $n \geq 3$ vertices such that $\deg(u) + \deg(v) \geq n$ for every pair of non-adjacent vertices u and v , then G has a Hamilton circuit.*

Important notes.

- Both theorems give *sufficient* conditions only — satisfying them guarantees a Hamilton circuit exists.
- They are *not necessary* — a graph may have a Hamilton circuit without satisfying either condition (e.g. C_5).
- Ore's theorem generalises Dirac's: if every vertex has degree $\geq n/2$, then for any non-adjacent u, v we get $\deg(u) + \deg(v) \geq n$.